



Transportation network and changes in urban structure: Evidence from the Jakarta Metropolitan Area

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ABSTRACT

The Jakarta Metropolitan Area (JMA) has a population of 28 million, a figure which is growing relative quickly. This study examines the role of transport infrastructure in explaining the suburbanization process at the community level in JMA. Aside from population density, it also uses night light intensity as proxy for economic activity. Using population census and night light data for 2000 and 2010, and the instrumental variable (IV) strategy, we found that improvements in highway and railway access promote population growth in city suburbs and non-city suburbs, respectively confirming the process of transportation-led suburbanization. In contrast, suburbanization of night light intensity was only evident in non-city suburbs and is likely to be driven by the transportation network stock.

1. Introduction

By 2015, four billion people were living in cities, accounting for 54% of the total world population; a growth of 73% from the 1990 urban population statistic (UN-Habitat, 2016). Rapid population growth and economic activity leads the expansion of cities and creates vast conurbation areas, comprising urban centres and suburbs. Internally, economic activities in cities expand beyond their administrative borders. More urban development is found in suburbs than in city areas, a phenomenon which is observed not only in cities in developed countries, such as Tokyo (Cybriwsky, 1993), but also in developing countries (Morichi & Acharya, 2013; Hassan, 2017).

Rapid suburb development is also evident in the Jakarta Metropolitan Area (JMA), the most populous and urbanized region in Indonesia. Consisting of 13 municipal areas,¹ JMA is home to 11% of Indonesia's population and contributes about 22% of the national gross domestic product (Indonesian Bureau of Statistics, 2010). Despite 14.5% increase in population, the proportion of people living in Jakarta decreased from 39.4% to 34.0% out of the total JMA population

between 2000 and 2010, indicating that more people were living in the suburbs. In addition, the urban population rose by 90% between 1990 and 2010 (Jones et al., 2016), transforming some outer areas of the JMA into new sub-centres. A similar trend was also seen with regard to economic activity, as industrial activity moved from the core of Jakarta to the suburbs (Henderson & Kuncoro, 1996).

The changes in the JMA population distribution are partly attributed to infrastructure expansion (such as the building of highways and railways). The development of highways and railway lines leads to suburb-urban commuting becoming more accessible. Recent transport infrastructure development also has the potential to promote the relocation of production activities to the fringes of Jakarta. A study by Henderson and Kuncoro (1996) suggests that there was some degree of suburbanization in the manufacturing sector from 1990 to 2000 in response to the development of a toll road running from Jakarta. Identifying the causal relationship between transport infrastructure and changes in the urban spatial structure of the Jakarta Metropolitan Area is crucial in examining the extent to which the urban structure-transportation network nexus is prevalent in the context of megacities in developing countries.

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¹ Jakarta consists of six municipalities, namely Central Jakarta, North Jakarta, East Jakarta, South Jakarta, West Jakarta, and Thousand Islands. A further seven municipalities are located in the surrounding suburbs, being Bekasi Regency, Bekasi City, Depok City, Bogor Regency, Bogor City, Tangerang Regency, and Tangerang City.

Our paper estimates the relationship between improvement in transportation access and changes in the urban structure of the JMA for a sample of 1483 communities (*kelurahan*) in years 2000 and 2010; these urban structure changes are reflected by changes in population density and economic activity. As current economic indicators are unable to measure economic activities at the community level, we use light intensity data as a proxy for economic activity. To address the potential simultaneity bias in the regression model, we used specifications that measured the improvement in transportation access using old transport infrastructure access i.e., access to the old railway network, old primary roads, and Anyer-Panarukan Road). Our results suggest that improvement in access to highways and access to railways station has promoted the suburbanization process in the JMA, although the roles differ.

Our study contributes to the body of literature relating to transportation network and urban structure, and the suburbanization process in particular, in two ways. A novel is that we propose night light as a proxy for the extent of the transport network in explaining the dynamics of economic activity. This expands our understanding of the fact that the suburbanization process is not only captured by faster population growth (Baum-Snow, 2007; Garcia-Lopez and Angel, 2012; Garcia-Lopez, Angel, Adelheid Holl, & Viladecans-Marsal, 2015), but is also explained by faster light night intensity growth in suburbs than in urban ones. Our study is also of international interest, as like other metropolitan areas in developing countries, JMA is experiencing a fast-changing spatial structure. The study is of further potential use in understanding the suburbanization process in Asian primate cities such as Jakarta. The results confirm that transportation-led suburbanization is a common phenomenon, not only in cities in developed countries, but also in developing countries.

The paper proceeds as follows. Section 2 explains the dynamics of urbanization and the urban transport infrastructure in JMA, while Section 3 presents the method and data. The results are presented in Section 4 and the final section provides the conclusions.

2. Context: the transport network and urban development in JMA

The last two decades have seen a rapid improvement in transport infrastructure, including both highways and railways, in JMA. This rapid development has improved the accessibility of nearby suburbs and expanded people's options in terms of relocating to such areas. The development of highways and railways has decreased the commuting distance from communities in JMA. The construction of the first section of the JMA highways was the Jagorawi Toll Road in 1973, which was completed in 1978 and has a total length of 59 km. The aim of the construction of this toll road was to improve mobility between Jakarta and Bogor. In 1984, the second section of the JMA highways, namely the Jakarta-Tangerang Toll Road, was opened, which connects the west side of Jakarta with Tangerang City. This section was later extended to Merak, Banten, with the objective of connecting JMA to Merak Port. Between 1987 and 2007, as many as 10 new highway sections were completed across JMA, with a total length of 178.2 km (Table 1 and Fig. 1). These new highway sections were mainly part of the Jakarta Outer Ring Road (JORR) project.

Improvement in the transport infrastructure of JMA has also occurred in the railways. In 1986, the only railway line in the area was the Jakarta – Bogor Line. Between 1986 and 2011, five new lines were opened across JMA (Table 2 and Fig. 1). These railway lines had already existed since the Dutch era, but the stations were not opened for commuters until 1987 (Jatinegara – Manggarai Loopline). The first of the JMA railway lines, the Jakarta – Bogor line, has a length of 54.8 km (the longest line in JMA) and connects North Jakarta with Bogor City, with 24 stations. The shortest line is the Duri – Tangerang line, which has operated since 1997, with a total length of 19.29 km, connecting West Jakarta with Tangerang City, with 11 stations. From 2000 to 2010, there was an increase in the number of daily trips, shown by a greater number of passenger trains that passed through several main

Table 1

Highway development in the JMA.

Source: Indonesia Highway Authority (2015)

Sections	Length (km)	Year
Cawang – Tomang – Grogol – Pluit	23.55	1987
Jakarta – Cikampek	83.00	1988
Cawang – Tj. Priuk – Ancol Timur – Pluit	27.05	1989
JORR Seksi S	14.25	1995
Serpong – Pondok Aren	7.25	1999
Pondok Aren – Ulujami	5.55	1999
JORR W2S	5.72	2005
JORR E1 Utara Sie.3	4.35	2006
JORR E3	3.75	2007
JORR E1 Utara Sie.4	3.75	2007
Total additional highways	178.22	

stations in JMA, such as Bogor station (from 156 to 170 trips), Depok (from 178 to 223 trips), Serpong (from 46 to 53 trips), Tangerang (from 20 to 29 trips), and Bekasi (from 85 to 108 trips).²

The development of the highway network has directly affected 567 communities in JMA, with them becoming 8.78 km closer to highway ramps between 2000 and 2010. Meanwhile, the opening of new railway lines has brought 942 communities closer to railway stations with an average change in distance of 12.69 km during the same period. Between 2000 and 2010, three new highway sections were built, two of which were part of the JORR project. The longest of these was the JORR East (JORR-E) section, which connects some parts of East Jakarta, North Jakarta, and Bekasi with the highway system of JMA. Another section completes the Jakarta Inner-Ring Road, namely the Kebon Jeruk-Penjaraningan section.

Improvements in the transport infrastructure since the late 1980s have gone hand-in-hand with the economic development process that has taken place in JMA. The rapid development of Jakarta and the surrounding regions has intensified during the last two decades, due to the agglomeration effect following the development of several industrial estates and residential towns (Hudalah, Dimitra, Tommy & Johan, 2013). In 1980, the population of JMA represented around 6 percent of the total Indonesia population. By 2010, this figure had increased to approximately 11%. The population density of JMA had reached 79.4 inhabitants per hectare by 2010, being particularly concentrated in Jakarta and the immediate surrounding suburbs. These figures represent a 14.8 person per hectare increase, or a 23% increase, over the 2000 figures (see Fig. 1). For comparison, in 2010 the national population density was 1.2 inhabitants per hectare, which had increased by 15.2% from 2000 (Indonesian Bureau of Statistics, 2000; 2010). The dense agglomeration process in JMA was also in line with the concentration of economic activities, as JMA produced about 23.6% of national GDP between 2000 and 2010 (Indonesian Bureau of Statistics, 2018). During the 2000s, the JMA economy grew by around 5.8% annually, outpacing the national growth rate of 5.2%. By 2016, the per capita GDP of JMA had reached 97.6 million rupiahs, or more than twice the national figure (Indonesian Bureau of Statistics, 2018). The middle-income class is growing in JMA, and by 2020, more than 21 percent of the national total will be residing in JMA (Rastogi, Eddy, Dean & Tunnee, 2013).

While the region is experiencing a strong urbanization process as a whole, the internal structure of JMA is characterised by a suburbanization process, indicated by the increasing population density in several regions near the border of Jakarta (Fig. 2). The process of urban sprawl is more evident in regions located near the transport infrastructure (highways and railways) and has occurred in Bogor, Depok, Tangerang and Bekasi (Bodetabek). New built-up areas have been more

² Data taken from PT. Kereta Commuter Jabodetabek.

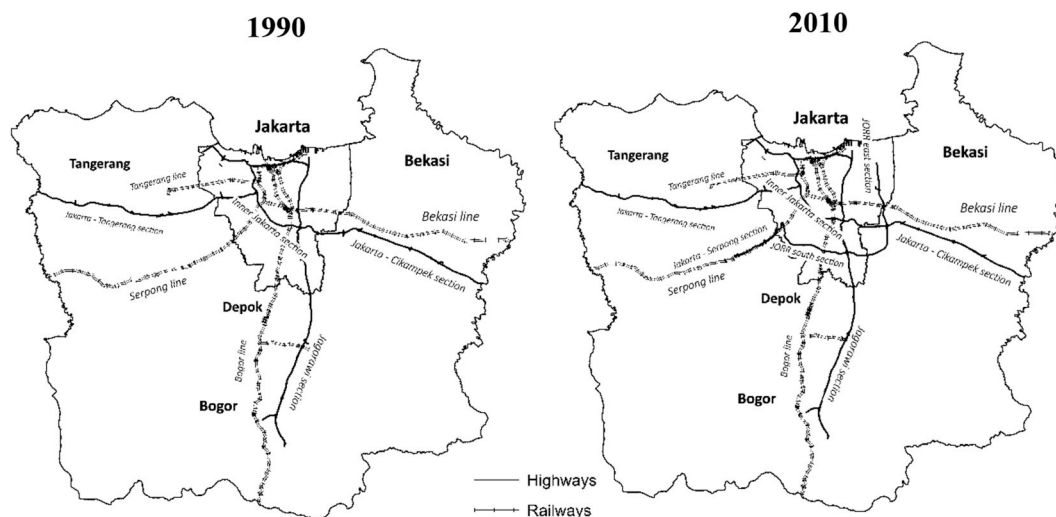


Fig. 1. Development of highway and railway networks, 1990-2010

Source: Indonesia Highway Authority (2015), PT. Kereta Api Indonesia (2016), PT. KAI Commuter Jabodetabek (2016).

Table 2

Railway development in the JMA.

Source: PT. Kereta Api Indonesia (2016), PT. KAI Commuter Jabodetabek (2016)

Railway Line	Length (km)	Year
Jatinegara – Manggarai Loopline	2.65	1987
Tanah Abang – Serpong Line	24.28	1992
Jakarta – Bekasi Line	27.34	1992
Duri – Tangerang Line	19.30	1997
Tanah Abang – Parung Panjang Line	34.55	2009

prevalent on the border of Depok city, Bogor regency and Bogor city. The population density of these areas increased to 50–150 inhabitants per hectare during the period 2000–2010 from less than 50 inhabitants per hectare in 2000. Between 2000 and 2010, the average growth in population density in non-city suburbs (39.0%) was also higher than in city suburbs (34.3%). The increase in population density was evident in the area around South Jakarta, Depok and Jakarta following the construction of the East Jakarta Outer Ring Road (JORR). As a result, 37% of urban land in JMA was in the core area of Jakarta by 2010, and it is expected that this proportion will decreased further, along with the

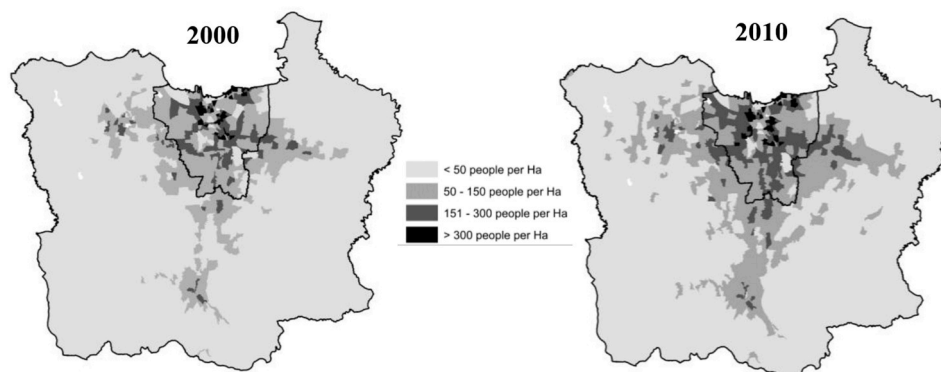


Fig. 2. Population density in the JMA

Source: Indonesian bureau of Stock and Yogo (2005, 2010).

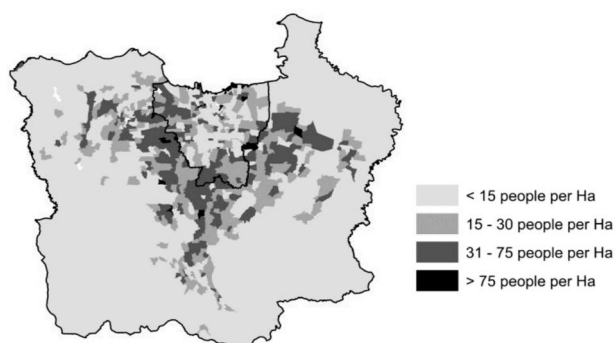


Fig. 3. Changes in population density 2000-2010

Source: Indonesian bureau of Stock and Yogo (2005, 2010).

higher population growth in the suburbs (see Fig. 3).

We also use night light data as a proxy for general economic activity to examine the dynamics of the JMA urban structure. The literature is increasingly using such an approach when real data on economic activity, such as GDP, are difficult to obtain (Henderson, Storeygard, and Weil 2012; Storeygard, 2016). Similar datasets have been used widely to estimate other variables, such as energy consumption (Elvidge et al., 1997; Letu et al., 2010; Zhao, Ghosh, & Samson, 2012) and population (Elvidge et al., 1997; Sutton, RobertsElvidge, & Baugh, 2001; Zhuo et al., 2009). Fig. 4 presents the dynamics of night light intensity in JMA between 2000 and 2010. The less-shaded areas represent the areas with high luminosity levels. Light coverage in 2010 was brighter in several areas in the JMA and covered a greater area than in 2000. During the period 2000–2010, light levels gradually increased, with the greatest increase concentrated in the suburbs adjacent to Jakarta

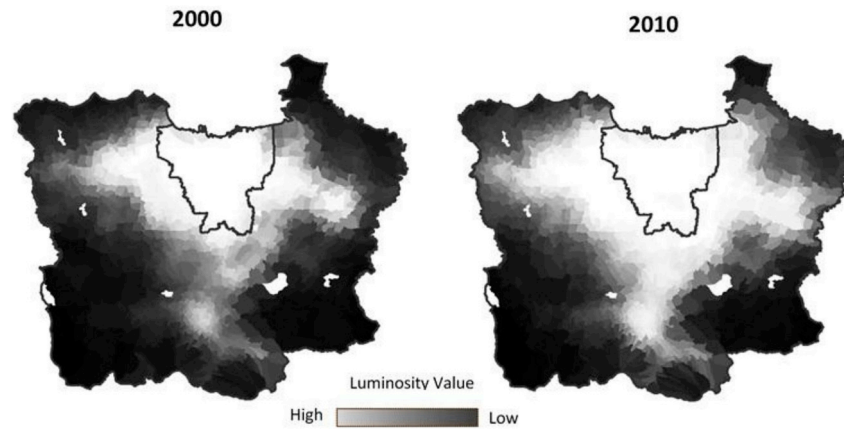


Fig. 4. JMA night time lightSource National Center s for Environmental Information (2016).

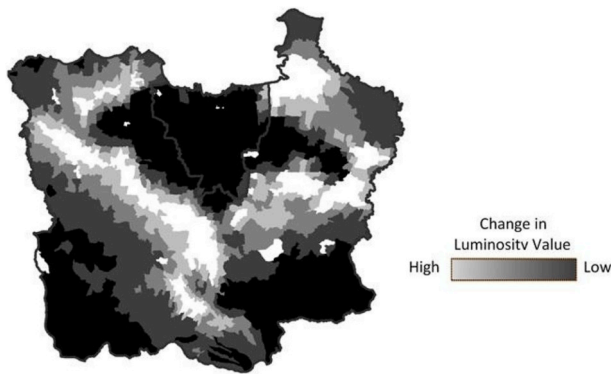


Fig. 5. Change in night time light 2000–2010 Source: National Centres for Environmental Information (2016).

(Fig. 5).

The combination of transport improvement, middle-income class expansion, and the population growth pattern in suburbs explains the shift in the mobility pattern of JMA residents. 73% of trips were made by private modes by 2010, almost double the figure of 2002 (JUTPI, 2012). The presence of highways connecting residential areas in the suburbs with Jakarta city centre encourages private-based commuting. There was a 50% increase in suburb-to-urban trips in JMA between 2002 and 2010 (JUTPI, 2012). Along with the extensive suburbanization process, it is also predicted that suburb-to-urban trips will grow by 146% by 2030, a much figure higher than the 104% growth in intra-urban trips in the same period (JUTPI, 2012).

3. Identification

3.1. Model and data

To estimate the impact of transportation network improvement on the suburbanization process, we adapt the regression model by Garcia-Lopez and Angel., 2012, as follows:

$$\Delta \ln(y_i) = \alpha + \gamma_1 \Delta r_i^{high} + \gamma_2 r_i^{rail} + \beta x_i + e_i, \quad (1)$$

where $\Delta \ln(y_i) = \ln(y_{i2010}) - \ln(y_{i2000})$ is growth of outcome variables at community i . This study uses two outcome variables at the community level to quantify the urban spatial structure, namely population density and night light intensity. We use two waves of population census data—years 2000 and 2010 (Indonesia Bureau of Statistics, 2000; 2010)—to generate population figures at the community level. The land

size is calculated from vector digital maps provided by the Geospatial Information Bureau of Indonesia.

We use the stable-light data obtained by the Defense Meteorological Satellite Program from the National Oceanic and Atmospheric Agency (NOAA) as a proxy for economic activity. The data has pixel values ranging from 0 to 63. We overlay the stable-light data and the Jakarta Metropolitan Area map at the community level. Combining both dataset, we generate a light intensity value for each community by averaging the value of pixels within each community. The stable-light data has resolution ranging from 0.55 to 2.77 km² for each pixel size.³ Since the average village area in the JMA is approximately 4.55 km², we believe the night light data remains representative at the community level.

$\Delta r_i^{high} = r_{i2010}^{high} - r_{i1990}^{high}$ measures improvement in access to transportation network highway ramp in kilometre. Meanwhile, r_i^{rail} measures the distance to closest railway station in 1990 to capture the frequency increase of train services along the line (Garcia-Lopez and Angel., 2012). The access variables are measured as the straight-line distance from the centre of a community to the nearest highway ramp or railway station. To obtain these variables, we use the same vector digital maps as for land size calculation, as the maps also provide information on the position of the highways and railways. During the period 1990–2010, 45 km of new highways were built, with 26 new ramps. Meanwhile, the railway network was extended by 81 km, with 22 new stations opened.

x_i represents the vector of control variables. We categorize the control variables according to three groups, namely distance, geography, and old population control variables. The distance control variables include distance to central Jakarta, distance to central district, and distance to coastal line. As were the transport variables, the distance control variables are obtained from the vector digital maps provided by the Geospatial Information Bureau of Indonesia. The geography control variables contain the ruggedness variable, land size, and elevation. Old population control variable includes 1961 population at the district level and was taken from Winarso (2010). Appendix A shows the summary statistics of our variables.

Our hypothesis is that improvement in transportation access, reflected by shorter distances to highway ramps or railway stations, will increase both population density and economic activity. Thus, the

³ The DMPS OLS instruments metaview. This page explains the satellites resolution at high resolution (0,55 km), and at low resolution (2,7 km) <https://www.ngdc.noaa.gov/metaview/page?xml=NOAA/NESDIS/NGDC/STP/DMSP/iso/xml/G01119.xml&view=getDataView&header=none>. accessed on 22/01/2018. Huang, Yang, Gao, Yang and Zhao (2014) found that the stable light images from DMSP satellites has spatial resolution of 1 km.

development of transportation access connecting suburbs may cause the suburbanization process in the JMA. Being closer to highway ramps or railway stations, indicated by shorter distances to the ramps/stations, decreases travel/commuting time for residents. Similarly, improved access decreases firms' logistical costs for receiving their inputs or delivering their products to consumers. If the access effect on suburbanization is present, we also expect that γ_j would be more elastic for suburbs than it would be for Jakarta. More elastic γ_j in suburbs indicates that any transport access improvement will lead to a greater increase of population density or night light intensity.

3.2. Endogeneity and estimation strategy

Estimating Equation (1) using the ordinary least square (OLS) will provide inconsistent distance parameters (γ_j) due to the reverse causality problem. Improvement in transportation infrastructure may be driven by increases in population density or economic activity that create demand for better transport infrastructure. Recent transportation network is also likely to acknowledge the presence of old infrastructures (Duranton & Turner, 2012; Garcia-Lopez et al., 2015; Yudhistira and Sofiyandi, 2018). This reasoning seems even truer for railways service as the network was developed on 1925 railways network built during period of Dutch colonialization. The 1990 distance to railway station also tends to be endogenous, as the decision to increase the frequency could have been affected by the increase in population density or economic activity near the stations. To overcome this endogeneity issue, the instrumental variables (IV) method is applied (Wooldridge, 2010). The idea is to find a set of variables that are strongly correlated with the variables, yet uncorrelated with the error term.

We propose three distance variables as potential IVs: distance to Anyer-Panarukan road, distance to old arterial road, and distance to old railway station. The Anyer-Panarukan road was constructed at the governorship of Herman Willem Daendels (1808–1811) of Dutch colonial era, stretching at the northern part of Java island (Nas, and Pratiwo, 2002). Our old arterial road extends the Anyer-Panarukan road with arterial roads built in early period of Indonesia's independence, such as Sukabumi road and Jakarta-Cirebon road. The old railway station was drawn from the station network built during the Nederlandsch Indische Spoorweg Maatschappij (NISM) era (van Ballegoijen de Jong, 1993). These variables represent the older transport access in the JMA. We argue that these variables must be considered by planners in developing the highway/railway system in the JMA since national regulations state that any new highways must consider the existing connectivity of the area. We also argue that these variables have no correlation with current population density and night light intensity.

4. Results

4.1. First stage and reduced form

We follow Murray (2006, 2010) when choosing the potential instrument variables. First, we estimate equations that predict transportation improvement (first-stage estimates) and the change in population density and night light intensity (reduced-form estimates) as a function of each instrument variable. Next, we select an instrument whose coefficient is significantly different from zero with a sign that justifies the following two assumptions: 1) positive or negative first-stage results indicate that transportation improvement is located close to or far from the instrument; and 2) negative reduced-form results indicate that proximity to the instrument accelerates the growth of population density and night light intensity. To avoid over identification problems, the

number of instruments assigned is equal to the number of endogenous variables on the right-hand side. Finally, we use Stock-Yogo criteria for the weak instruments test for any pair of potential instruments. The null hypothesis of the weak instruments is rejected if the Kleibergen-Paap F-statistic on an instrument exceeds the size and relative bias critical value (Stock & Yogo, 2005).

The first-stage and reduced-form estimates are presented in Table 3. Panel A shows the first-stage results for the changes in transportation improvement in terms of the distance to the nearest highway ramp (shown in the odd columns) and the distance to the nearest railroad (shown in the even columns) that took place in the whole of JMA, Jakarta, the city suburbs, and other suburbs. Panel B displays the reduced-form results of population density and night light intensity.

Result in Table 3 shows that our instruments perform relatively well in the first-stage regression. In most specifications, all instruments are statistically significant in explaining the variation of the improvement in distance to highway ramp and distance to railway station. Nevertheless, the result also suggests that the IV strategy does not perfectly address endogeneity problem from the OLS. Our choice of instruments may leave some biases as reduced-form result shows some instruments are not statistically significant. For instance, estimates of distance to old railway in the Jakarta and city suburbs samples are not statistically different from zero. Our 2SLS estimates may still suffer from some biases, and hence the correction made by the IV approach should be interpreted cautiously.

Aside from the problem of imperfect instruments, our first-stage and reduced-form regressions further indicate that the choice of instruments may vary across samples. We use access to the nearest old railway station as a potential instrument for all specifications as its first-stage coefficients are statistically different from zero in explaining both endogenous variables in all the samples. We choose distance to the Anyer-Panarukan road as the second instrument for the JMA sample since it is statistically significant in explaining the endogenous variables. As for the Jakarta sample, we prefer distance to the Anyer-Panarukan road as the second instrument since it provides a better Kleibergen-Paap F-statistic than the old primary road for population density and night light intensity. As for the city suburbs, distance to the Anyer-Panarukan road and old primary road are both potential as second instrument. We check both instruments using the Kleibergen-Paap F-statistic and find that distance to the Anyer-Panarukan and old primary road are more suitable instruments for population density and night light intensity specifications, respectively. Similarly, distance to the old primary and Anyer-Panarukan road are more appropriate as second instrument for population density and night light intensity specifications for other suburbs sample, respectively.

4.2. 2SLS results for population density

Table 4 presents our estimation results for Equation (1). Panel A shows the effect of changes in log population density for Jakarta area, while Panels B and C represent the effect for city suburbs and other suburbs, respectively. We lump results for the JMA in the Appendix B.

We start our analysis from OLS result presented in Column 1. In general, we find that highway and railway system tend to reduce the population growth in the city centre of the JMA. The coefficients for the Jakarta are positive and statistically different from zero. Coefficient for highway indicate that improvement in access to nearest highway ramp had an adverse impact on population density growth in the Jakarta. Meanwhile, the estimates for city suburbs and other suburbs are statistically not different from zero; the presence of highway system does not hasten the growth of population density in suburbs. Similarly, coefficient on railway shows that the density in communities of Jakarta located further away from the station will grow faster. Similar effect is

Table 3
First-stage regressions and reduced form, OLS.

A. First stage regressions								
First-Stage Regressions								
	JMA		Jakarta		City suburbs		Other suburbs	
	Δ distance to highway ramp	distance to railway station	Δ distance to highway ramp	distance to railway station	Δ distance to highway ramp	distance to railway station	Δ distance to highway ramp	distance to railway station
Anyer-Panarukan road	−0.053* (0.008)	−0.137* (0.019)	−0.356* (0.043)	−0.025 (0.063)	−0.373* (0.039)	0.185* (0.042)	−0.049* (0.010)	−0.300* (0.026)
Old primary road	0.007 (0.015)	0.295* (0.051)	0.318*** (0.169)	0.325 (0.202)	0.565* (0.125)	−0.364** (0.161)	−0.004 (0.015)	0.338* (0.053)
Old railway	0.073* (0.013)	0.710* (0.047)	0.168* (0.063)	0.492* (0.097)	−0.296** (0.116)	0.587* (0.104)	0.047* (0.014)	0.439* (0.054)
ln (1961 district population)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographical variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1472	1472	261	261	340	340	871	871
Adjusted R-sq	0.180	0.857	0.507	0.729	0.537	0.896	0.250	0.802
B. Reduced Form								
	Reduced Form JMA		Jakarta		City suburbs		Other suburbs	
	Δ ln(population density)	Δ ln(night light)	Δ ln(population density)	Δ ln(night light)	Δ ln(population density)	Δ ln(night light)	Δ ln(population density)	Δ ln(night light)
Anyer-Panarukan road	−0.001 (0.001)	0.003* (0.0006)	−0.004 (0.003)	0.0001 (0.0003)	0.007*** (0.004)	−0.0009** (0.0005)	−0.0009 (0.001)	0.003* (0.0007)
Old primary road	−0.013* (0.002)	−0.019* (0.002)	0.075* (0.018)	0.002** (0.001)	0.037* (0.014)	0.006* (0.002)	−0.007* (0.002)	−0.009* (0.002)
Old railway	−0.004* (0.002)	−0.009* (0.001)	0.001 (0.006)	0.0001 (0.000)	−0.004 (0.009)	0.00002 (0.001)	0.0005 (0.002)	−0.003* (0.001)
ln (1961 district population)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographical variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1472	1472	261	261	340	340	871	871
Adjusted R-sq	0.220	0.692	0.426	0.926	0.398	0.971	0.195	0.454

Note: *robust standard error* in parentheses. ***, **, * indicate statistical significance at 10%, 5%, and 1%. Geographical variables include the ruggedness variable, land size, and elevation. Distance variables include the distance to central Jakarta, distance to central district, and distance to coastal line.

also found for the city suburbs, but the estimate was much smaller than it was Jakarta.

The OLS result, however, is suspected to be invalid due to endogeneity problem. Columns 2, 3, 4, and 5 of Table 4 provides two-stage least squares (2SLS) estimates. Column 2 includes transportation variables and lagged population density. Column 3 adds district population at 1961. Columns 4 and 5 add geographical and distance variables, respectively.

Our 2SLS result on population density confirm our hypothesis that improvement in transport infrastructure contributes to the change in urban structure in the Jakarta Metropolitan Area. However, the pattern was relatively different from its OLS counterparts. In addition, the effects differed according to the type of transport infrastructure. There is a tendency of suburbanization led by highways is concentrated in cities located in suburbs. Coefficients on highways show that communities with better highway access increase in population density more quickly in city suburbs. A kilometre improvement of highway access in city of

suburbs corresponds to 3.7–4.8% growth in population density, depending on the set of control variables. However, although the adverse effect was seen in Jakarta and other suburbs, it was only statistically significant for other suburbs' estimates. Increased improvement in terms of highway access resulted in lower population density growth in those areas.

Our estimates for railways show that the improvement in railway access is also important for residents' decision to live in other suburbs. We found a positive effect that was statistically different from zero for other suburbs only. A kilometre closer to railway station in other suburbs corresponds to 1.3–1.7% growth in population density. The estimates for the Jakarta and city suburbs are statistically insignificant under full specification model (column 5). This result can be explained as railway service is more reliable and saves more travel time for longer distance travel, promoting higher population growth in the other suburbs.

Table 4
2SLS regression results for population density.

	$\Delta \ln(\text{population density})$				
	OLS		IV		
	(1)	(2)	(3)	(4)	(5)
Panel A: Jakarta. Instruments: old railway and Anyer-Panarukan road					
Δ distance to highway ramp	0.009*** (0.005)	0.050* (0.016)	0.050* (0.016)	0.015 (0.011)	0.013 (0.010)
distance to railway station	0.011* (0.004)	0.042* (0.009)	0.042* (0.009)	0.023* (0.008)	0.014 (0.012)
$\ln(2000 \text{ population density})$	−0.049** (0.019)	−0.051** (0.021)	−0.051** (0.021)	−0.038*** (0.021)	−0.047** (0.020)
$\ln(1961 \text{ district population})$	Yes	No	Yes	Yes	Yes
Geographical variables	Yes	No	No	Yes	Yes
Distance variables	Yes	No	No	No	Yes
Observations	261	261	261	261	261
K-P-F statistic on instruments		21.44	21.44	28.84	10.71
Adjusted R-sq	0.383				
Panel B: City suburbs. Instruments: old railway and Anyer-Panarukan road					
Δ distance to highway ramp	0.003 (0.006)	−0.037** (0.015)	−0.008 (0.154)	−0.04*** (0.024)	−0.048** (0.019)
distance to railway station	0.006 (0.006)	−0.004 (0.019)	−0.036 (0.203)	0.012 (0.039)	−0.031 (0.024)
$\ln(2000 \text{ population density})$	−0.188* (0.025)	−0.171* (0.017)	−0.241 (0.405)	−0.122 (0.122)	−0.178* (0.033)
$\ln(1961 \text{ district population})$	Yes	No	Yes	Yes	Yes
Geographical variables	Yes	No	No	Yes	Yes
Distance variables	Yes	No	No	No	Yes
Observations	340	340	340	340	340
K-P-F statistic on instruments		2.299	0.022	0.86	11.26
Adjusted R-sq	0.369				
Panel C: other suburbs. Instruments: old railway and old primary road					
Δ distance to highway ramp	0.003 (0.004)	0.206* (0.047)	0.089* (0.025)	0.145* (0.037)	0.150* (0.051)
distance to railway station	0.003*** (0.001)	−0.022* (0.004)	−0.017* (0.002)	−0.020* (0.003)	−0.014*** (0.009)
$\ln(2000 \text{ population density})$	0.036 (0.027)	−0.120* (0.038)	−0.054** (0.021)	−0.067*** (0.034)	−0.067 (0.044)
$\ln(1961 \text{ district population})$	Yes	No	Yes	Yes	Yes
Geographical variables	Yes	No	No	Yes	Yes
Distance variables	Yes	No	No	No	Yes
Observations	871	871	871	871	871
K-P-F statistic on instruments		17.78	34.84	18.64	9.121
Adjusted R-sq	0.186				

Note: *robust standard error* in parentheses. ***, **, * indicate statistical significance at 10%, 5%, and 1%. Geographical variables include the ruggedness variable, land size, and elevation. Distance variables include the distance to central Jakarta, distance to central district, and distance to coastal line. K-P F-statistic corresponds to Kleibergen-Paap F-statistic for weak instrument/s identification test. The null of weak instruments is rejected if the K-P F-statistic on the instrument/s passes Stock-Yogo critical value.

4.3. 2SLS results for night light intensity

Table 5 reports the estimates of Equation (1) for night light intensity. 2SLS estimation result is less intuitive, while we find strong

evidence of suburbanization of the economic activity using OLS estimation. The estimates are statistically different from zero and were only robust for several specifications in other suburbs. Better access to railway station tended to promote greater night light growth in the

Table 5
2SLS regression results for night light intensity.

	$\Delta \ln(\text{night light})$				
	OLS		IV		
	(1)	(2)	(3)	(4)	(5)
Panel A: Jakarta. Instruments: old railway and Anyer-Panarukan road					
Δ distance to highway ramp	−0.0009* (0.000)	0.001 (0.002)	0.001 (0.002)	0.0008 (0.001)	−0.0003 (0.001)
distance to railway station	0.001* (0.0003)	−0.0001 (0.0005)	−0.0001 (0.0005)	0.0002 (0.0006)	0.0007 (0.0007)
$\ln(2000 \text{ night light intensity})$	−0.769* (0.022)	−0.739* (0.035)	−0.739* (0.035)	−0.741* (0.031)	−0.774* (0.024)
$\ln(1961 \text{ district population})$	Yes	No	Yes	Yes	Yes
Geographical variables	Yes	No	No	Yes	Yes
Distance variables	Yes	No	No	No	Yes
N	261	261	261	261	261
K-P-F statistic on instruments		22.36	22.36	26.95	9.20
Adjusted R-sq	0.936				
Panel B: City suburbs. Instruments: old railway and old primary road					
Δ distance to highway ramp	−0.002* (0.001)	−0.007* (0.002)	0.0001 (0.006)	0.001 (0.003)	0.002 (0.002)
distance to railway station	−0.004* (0.001)	−0.003 (0.002)	−0.010 (0.011)	−0.007 (0.005)	0.001 (0.003)
$\ln(2000 \text{ night light intensity})$	−0.749* (0.020)	−0.668* (0.037)	−0.711* (0.029)	−0.811* (0.037)	−0.767* (0.021)
$\ln(1961 \text{ district population})$	Yes	No	Yes	Yes	Yes
Geographical variables	Yes	No	No	Yes	Yes
Distance variables	Yes	No	No	No	Yes
Observations	340	340	340	340	340
K-P-F statistic on instruments	–	7.701	0.506	0.929	7.602
Adjusted R-sq	0.973				
Panel C: other suburbs. Instruments: old railway and Anyer-Panarukan road					
Δ distance to highway ramp	−0.023* (0.002)	0.102* (0.023)	0.081* (0.020)	0.179* (0.046)	0.131** (0.063)
distance to railway station	−0.005* (0.001)	−0.013* (0.002)	−0.025* (0.002)	−0.031* (0.005)	−0.017** (0.007)
$\ln(2000 \text{ night light intensity})$	−0.300* (0.016)	−0.165* (0.021)	−0.340* (0.029)	−0.427* (0.048)	−0.417* (0.060)
$\ln(1961 \text{ district population})$	Yes	No	Yes	Yes	Yes
Geographical variables	Yes	No	No	Yes	Yes
Distance variables	Yes	No	No	No	Yes
Observations	871	882	882	882	882
K-P-F statistic on instruments		44.47	47.87	14.94	4.837
Adjusted R-sq	0.492				

Note: *robust standard error* in parentheses. ***, **, * indicate statistical significance at 10%, 5%, and 1%. Geographical variables include the ruggedness variable, land size, and elevation. Distance variables include the distance to central Jakarta, distance to central district, and distance to coastal line. K-P F-statistic corresponds to Kleibergen-Paap F-statistic for weak instrument/s identification test. The null of weak instruments is rejected if the K-P F-statistic on the instrument/s passes Stock-Yogo critical value.

other suburbs. On the contrary, the estimates suggest negative effect of highway improvement the other suburbs.

Table 6 explores the role of transportation stock on night light intensity growth as we argue that the growth may be better explained by the stock than the improvement. We estimate the effect of distance in 2000 distance to the nearest highway ramp and to the nearest railway

station. The dependent variable remains 2000–2010 changes in log night light intensity. Odd and even columns represent OLS and 2SLS estimates, respectively. Columns 1–2, 3–4, and 5–6 are estimates for the Jakarta, city suburbs, and non-city suburbs. In general, we find 2SLS and OLS results are relatively similar in coefficient sign but differ in magnitude and significance. The results support previous findings and

Table 6
Alternative specification for night light intensity.

	Jakarta		City suburbs		Other suburbs	
	IV	OLS	IV	OLS	IV	OLS
	(1)	(2)	(3)	(4)	(5)	(6)
2000 distance to highway ramp	0.0003 (0.001)	0.001** (0.0004)	0.012*** (0.006)	0.001*** (0.0008)	−0.009* (0.002)	−0.003** (0.001)
2000 distance to railway station	0.0005 (0.0006)	0.0005 (0.0004)	−0.003 (0.002)	0.0005 (0.001)	−0.005* (0.001)	−0.007* (0.001)
ln(2000 night light intensity)	−0.769* (0.046)	−0.757* (0.026)	−0.651* (0.061)	−0.744* (0.022)	−0.391* (0.021)	−0.349* (0.020)
ln (1961 district population)	Yes	Yes	Yes	Yes	Yes	Yes
Geographical variables	Yes	Yes	Yes	Yes	Yes	Yes
Distance variables	Yes	Yes	Yes	Yes	Yes	Yes
Instruments:						
Old railway	Yes		Yes		Yes	
Anyer-Panarukan roda	Yes		Yes		No	
Old primary road	No		No		Yes	
Observations	261	261	340	340	882	871
K-P-F statistic on instruments	7.755		4.777		274.6	
Adjusted R-sq		0.926		0.971		0.454

Note: *robust standard error* in parentheses. ***, **, * indicate statistical significance at 10%, 5%, and 1%. Geographical variables include the ruggedness variable, land size, and elevation. Distance variables include the distance to central Jakarta, distance to central district, and distance to coastal line. K-P F-statistic corresponds to Kleibergen-Paap F-statistic for weak instrument/s identification test. The null of weak instruments is rejected if the K-P F-statistic on the instrument/s passes Stock-Yogo critical value.

even suggest stronger evidence of the suburbanization in non-city suburbs. Better access to 2000 highway ramps fosters the night light growth. This result is supported by the fact that many industrial parks were developed in other suburbs, such as in Bekasi regency, exploiting the Jakarta-Cikampek highways section (Hudalah & Firman, 2012). Combining this the finding in the Table 5, we conclude that the highway access will accelerate economic activities in communities near the ramp. However, as the economic activities near the ramp are so dense, any improvement in highway access would prompt more economic activities to be located further away from the ramps.

4.4. Comparing OLS and 2SLS estimates

The considerable difference between the OLS and 2SLS estimation results further suggests that improvement to the transport infrastructure in the JMA is endogenous (Garcia-Lopez et al., 2015). In most cases, the 2SLS estimates were larger than were the similar OLS estimates. For example, in Panel A of Table 4, at 0.01, this 2SLS estimate for the highway ramp is higher than is the corresponding OLS estimate of 0.009. In Panel B and Panel C, the difference in magnitude between the 2SLS and OLS estimates are fairly consistent. However, some 2SLS estimates have a negative coefficient sign.

Among various potential sources of difference discussed by Duranton and Turner (2012), we argue that reverse causation problem can explain better these results. Communities where the new roads and train stations were built likely to have relatively low population density, as well as low population growth, with a small number of economy activities. Instead of the near-city centre location, the government prefers to build infrastructure in such regions due to a lower cost of land acquisition, e.g. new highways are built by acknowledging the presence of old transport infrastructure. Such preference seems even truer for

railways service as the new stations in Serpong and Bekasi Line were mainly opened in less developed communities. As a result, those regions experience a better effect of improvement in transportation access than more developed regions. A similar reasoning also applies for the difference of estimates of the effect of highway ramp and railway infrastructure proximities on the change of night light intensity. As the transport infrastructure are built in the newly opened location, more people and business activities will move into such location and therefore implies higher electricity use during the night time.

4.5. Relating the estimates with prior studies

We provide empirical evidence of the role of transportation infrastructure in causing suburbanization in the JMA. Previous studies on the urban structure of the JMA only provide descriptive analysis on the transportation-urban structure nexus of the JMA (Firman, 2004; Henderson & Kuncoro, 1996; Hudalah & Firman, 2012). Our estimates provide evidence that improvements in transport access as well as the initial condition of transport access promotes suburbanization of both population and economic activity. Specifically, two main findings are provided. Firstly, improvement in highways and better access to railways led the suburbanization process in such a way that it promoted greater population density growth and night light intensity in suburbs, although the roles differ. Improvement in highways access promotes more people to live in city suburbs, while railway access prompted more people to live in other suburbs. Second, suburbanization process of economic activity, represented by night light intensity, is more evident in other suburbs and rather driven by the stock of the transport infrastructure.

Our study adds to the literature on infrastructure and suburbanization for the case of a city in an emerging country. Despite the

existing understanding of the transportation-urban structure nexus, data availability issues have meant that empirical evidence has only recently been tested. The current literature shows that railways and highways contribute to urban growth (Atack, Bateman, Haines, & Margo, 2009; Duranton & Turner, 2012), land use change (Funderburg et al., 2010), productivity (Datta, 2012; Holl, 2016), drive suburbanization and decentralization (Baum-Snow, 2007; Garcia-Lopez and Angel, 2012; Garcia-Lopez, Holl, and Viladecans-Marsal 2015; Baum-Snow, Brandt, Vernon Henderson, Turner, & Zhang, 2015; Garcia-López, Hémet, and Viladecans-Marsal 2017), and subcentre formation (Garcia-López, Hémet, and Viladecans-Marsal 2017).

Our estimates for the population were relatively small compared to the range reported by Garcia-Lopez and Angel. (2012) for highways. Nevertheless, our estimates on highways are consistent with other studies. An additional highway passing city centres has been found to reduce their population by 18% in the USA (Baum-Snow, 2007), 8–9% in Spain (Garcia-Lopez, Holl, & Viladecans-Marsal, 2015), and at least 5% in China (Baum-Snow et al., 2015). Furthermore, our estimates for railways are rather weak in explaining JMA's suburbanization process, in contrast with 4% increase in population density for each km closer to a railway station in Paris (Garcia-López, Hémet, and Viladecans-Marsal 2017).

We did not find any closely comparable studies using night light intensity as proxy for economic activity and a similar estimation strategy to explain the suburbanization process. Nonetheless, our results are in line with a study by Baum-Snow et al. (2015), which finds that radial railroads displaced industrial GDP as well as employment in city centres.

5. Conclusions

As one of the biggest urbanized areas in South East Asia, the Jakarta

Metropolitan Area (JMA) has experienced dynamic changes in its urban spatial structure. This study investigated the extent to which the changes were driven by transportation infrastructure development in the JMA using population data and village-level data from the years 2000 and 2010. The novelty of this study lies in using change in night light intensity as proxy for economic activity as an alternative variable outcome of suburbanization, aside from changes in population density, which have been presented by other studies.

By using the instrumental variables approach, we find that highway and railway networks can explain the changes in urban spatial structure in the JMA. Improvement in highway access leads growth of population in city suburbs. In contrast, proximity to the stock of highway access explains why the night light as an alternative proxy of economic activity has positive growth in other suburb areas. Meanwhile, access to railways promotes population and night light growth in other suburbs.

The contribution of this study is relevant as the results provide empirical evidence for the case of a typical young primary city in an Asian country. Our analysis is centred on the Jakarta Metropolitan Area, the primary city in Indonesia, which is characterised by a disproportionately larger population and better transportation infrastructure than other urban areas in Indonesia (Morichi & Acharya, 2013). In this case, we found that the improvement of transportation promotes the sprawling process, the source of which is arguably immigration from other provinces. Thus, finding policy packages to balance the sprawl effect and increasing residents' mobility through improved transportation will be relevant to the local government of the JMA.

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Appendix A. Descriptive Statistics for All Variables

Table A
Descriptive statistics for all variables.

	JMA		Jakarta		City suburbs		Other suburbs	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Change in								
population density (people/hectare)	14.8	19.6	18.9	24.7	27.1	24.0	8.9	11.9
night light intensity	7.8	7.4	0.3	1.4	6.9	7.5	10.5	6.8
distance to ramp (kilometers)	−0.8	2.2	−0.7	1.6	−1.6	3.0	−0.6	1.8
Distance to railway station (kilometres)	17.7	13.1	3.3	2.8	11.0	6.6	24.6	11.9
1961 population at district level (000 people)	1409.5	754.1	2905.0	0.0	1055.4	316.7	1103.4	337.4
Distance to coastal line (kilometers)	26.0	15.7	10.5	6.4	26.5	13.3	30.4	15.7
Distance to central Jakarta (kilometers)	31.8	15.9	8.4	4.8	26.0	11.1	40.9	10.7
Distance to central district (kilometers)	11.9	8.9	4.6	2.4	5.0	2.5	13.7	6.7
Ruggedness index	19.3	56.8	1.2	2.2	4.8	5.8	30.2	71.5
Area size (hectare)	455.1	469.6	245.3	205.4	252.1	160.2	595.5	547.8
Elevation (meter)	116.1	200.5	18.0	10.7	78.7	94.6	159.5	242.7
Distance to old railway station (kilometers)	10.8	8.7	2.5	1.9	4.8	2.6	15.5	8.1
Distance to Anyer-Panarukan road (kilometers)	11.2	10.2	3.5	3.1	6.5	5.2	15.3	10.8
Distance to old primary road (kilometers)	4.9	6.0	0.7	0.6	1.2	1.1	7.5	6.4

Appendix B. 2SLS results for the JMA

Table B presents OLS and 2SLS results for the JMA. Focusing on columns 4 and 8, we find improvement of highway access is associated with lower population density and night light intensity growth, respectively. In contrast, better railway access increases population density and night light intensity growth, respectively. The results for night light intensity also provide early evidence on suburbanization process since quicker growth happens in areas far away from the central Jakarta.

Table B
Changes in population density and night light intensity of the JMA.

	$\Delta \ln(\text{population density})$			$\Delta \ln(\text{night light})$		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Instruments: old railway and Anyer-Panarukan road						
Δ distance to highway ramp	−0.006*** (0.003)	0.048** (0.020)	0.082* (0.027)	−0.009* (0.002)	−0.020 (0.012)	0.020 (0.016)
distance to railway station	−0.003* (0.001)	−0.007* (0.003)	−0.013* (0.004)	−0.007* (0.001)	−0.005* (0.002)	−0.012* (0.003)
$\ln(2000 \text{ population density})$	−0.043* (0.013)	−0.097* (0.014)	−0.078* (0.018)			
$\ln(2000 \text{ night light intensity})$				−0.340* (0.013)	−0.348* (0.016)	−0.359* (0.017)
distance to central Jakarta	−0.003** (0.001)	−0.008* (0.002)	−0.005** (0.002)	0.006* (0.001)	0.003*** (0.002)	0.006* (0.002)
$\ln(1961 \text{ district population})$	Yes	Yes	Yes	Yes	Yes	Yes
Geographical variables	Yes	No	Yes	Yes	No	Yes
Distance variables	Yes	No	Yes	Yes		Yes
Observations	1472	1472	1472	1483	1483	1483
K-P-F statistic on instruments		31.2	22.99		30.7	19.95
Adjusted R-sq	0.196			0.669		

Note: robust standard error in parentheses. ***, **, * indicate statistical significance at 10%, 5%, and 1%. Distance variables include distance to central district and distance to coastal line. Geographical variables include the ruggedness variable, land size, and elevation. K-P F-statistic corresponds to Kleibergen-Paap F-statistic for weak instrument/s identification test. The null of weak instruments is rejected if the K-P F-statistic on the instrument/s passes Stock-Yogo critical value.

Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.retrec.2018.12.003>.

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